

(Answer any five, Symbols have their usual meaning)

1. (a) Compare between metal, semiconductor and insulator concerning their band diagram. 4
- (b) Explain electron-hole pair generation in intrinsic Si semiconductor at room temperature. Can it behave as an insulator at room temperature? Explain 3+2
- (c) How n-type semiconductor is formed? 3
- (d) Explain why in n-type semiconductor the Fermi level energy E_F is close to the conduction band edge E_c of the band diagram. 4
2. (a) Define Fermi-level. 2
- (b) Explain the rectangular shape of Fermi function at 0K. 6
- (c) Show that Fermi function is symmetrical about E_F i.e. $f(E_F + \Delta E) = 1 - f(E_F - \Delta E)$ 8
3. (a) Show that equilibrium concentrations of electrons (n_0) in a semiconductor can be expressed as :- 5
- (b) Show that intrinsic carrier concentration under equilibrium is constant and can be expressed as :- $n_i = \sqrt{N_C N_V} e^{-\frac{E_g}{2KT}}$. 5
- (c) Explain the formation of depletion region and barrier potential V_0 in a p-n junction diode under equilibrium. 6
4. (a) Explain with diagram the forward bias and reverse bias characteristics of p-n junction diode. 4+4
- (b) Write down Shockley diode equation for p-n junction diode. 2
- (c) Explain the avalanche and Zener breakdown mechanism in a p-n junction diode. 3+3
5. (a) Draw and explain the output waveform of the circuit shown in Fig. 1. 5
- (b) Draw and explain the output waveform of the circuit shown in Fig. 2. 5
- (c) What will be the output waveform of the circuit shown in Fig. 3. Explain. 6

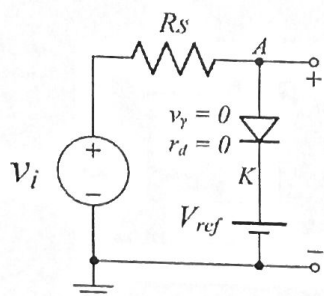


Fig. 1

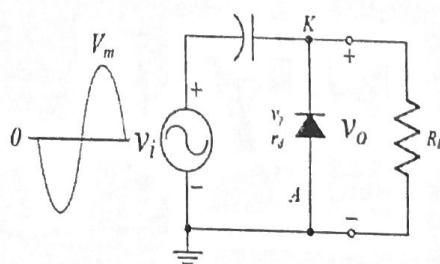


Fig 2

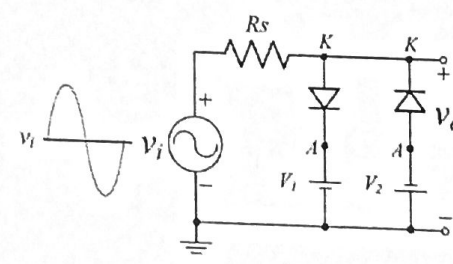


Fig.3

6. (a) Explain the operation of full wave rectifier circuit using bridge network. 6
- (b) Show that the output DC voltage (V_{dc}) and ripple factor of full wave rectifier output is: (i) $V_{dc} = \frac{2V_m}{\pi}$ (ii) $V_{rms} = \frac{V_m}{\sqrt{2}}$ (ii) $\Gamma = 48.3\%$ 3+3+4
7. (a) Explain different current components in a p-n-p transistor in active region with a suitable diagram. 5
- (b) Establish the relation between current amplification factors α and β . 5
- (c) Briefly explain the CB, CE and CC mode of operation in a transistor. 6
8. (a) What are the ideal characteristics of OPAMP? 4
- (b) Derive an expression of voltage gain of a non-inverting amplifier. 6
- (c) Draw the circuit diagram of a differentiator and find its output voltage expression. 6

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| 9. (a) | Draw and explain the operation of OP-AMP as an inverting amplifier. | 4 |
| (b) | What is the significance of the virtual ground that exists at the input of the ideal amplifier? | 2 |
| (c) | What is offset null in IC741? Design and analyse an integrator using an OP-AMP. | 2+4 |
| (d) | For an OP-AMP having a slew rate, of $2\text{V}/\mu\text{S}$, what is the maximum voltage gain that can be used when the input signal varies by 0.5V in $10\mu\text{S}$? | 4 |
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| 10.(a) | Explain the construction and formation of inversion layer in n-channel MOSFET with suitable diagram. | 8 |
| (b) | Draw and explain the output characteristics of MOSFET. | 8 |